

Development of an alignment method to allow intercomparison of environmental microplastic measures

Nela DOBIASOVA

First year of master's degree internship
as part of The Climate System course
Université Grenoble Alpes
Academic year 2024-2025

In collaboration with
Observatoire des Sciences de l'Univers de Grenoble
Institute des Geosciences et de l'environnement
CHIANTI team

from 5 May to 31 July 2025
under the supervision of Ian HOUGH

Abstract

Plastic pollution is a global issue impacting worldwide ecosystems and human and animal well-being and health. In order to effectively evaluate plastic waste management and global policies aimed at reducing microplastic (MP) pollution, it is essential to have a better understanding of how microplastics behave in the environment. In recent years, numerous protocols have been developed to quantify MP particles of different sizes in various ecosystems. This multiplicity of methods makes it difficult to compare and synthesize results from different studies. As part of my internship, I developed an approach based on cumulative particle counts that allows for the intercomparison of MP observational studies by extrapolating observed concentrations to any desired particle size range and estimating the associated uncertainty. The method was tested using both simulated and real datasets.

La pollution plastique est un enjeu global affectant les écosystèmes du monde entier et la santé et le bien-être humain et animal. Afin d'évaluer effectivement la gestion des déchets plastiques et les politiques mise en place pour réduire la pollution en microplastiques (MP), une compréhension approfondie du comportement des MP dans l'environnement est indispensable. Au cours des dernières années, divers protocoles d'observation ont été élaborés afin de quantifier les concentrations des MP dans les écosystèmes. Pendant mon stage, j'ai développé une méthode, basée sur la somme cumulative des particules, qui permet d'intercomparer les concentrations des MP reportées par les études d'observation, en extrapolant les concentrations observées et qui permet estimer l'incertitude associée. La méthode a été évaluée sur les données simulées ainsi que sur les données réelles.

Déclaration anti-plagiat *Antiplagiarism statement*

Document à **scanner** après signature et à **intégrer** au mémoire
Document to be scanned after signature and to be joined to the master thesis

Déclaration / *Statement*

1. Ce travail est le fruit d'un travail personnel et constitue un document original.
This paper is the result of personal work and is an original document.
2. Je sais que prétendre être l'auteur d'un travail écrit par une autre personne est une pratique sévèrement sanctionnée par la loi.
I know that claiming authorship of a work written by another person is a practice that is severely punished by law.
3. Personne d'autre que moi n'a le droit de faire valoir ce travail, en totalité ou en partie, comme le sien.
No one but me has the right to claim this work, in whole or in part, as their own.
4. Les propos repris mot à mot à d'autres auteurs figurent entre guillemets (citations).
Arguments taken over identically from other authors appear in quotation marks.
5. Les écrits sur lesquels je m'appuie dans ce mémoire sont systématiquement référencés selon un système de renvoi bibliographique clair et précis.
The texts on which I rely in this master thesis are systematically referenced according to a clear and precise bibliographic system.
6. Déclare avoir obtenu les autorisations nécessaires pour la reproduction d'images, d'extraits, figures ou tableaux empruntés à d'autres œuvres.
I declare that I have obtained the necessary authorizations for the reproduction of images, extracts, figures or tables borrowed from other works.

NOM/*LAST NAME* : DOBIASOVA

PRENOM/*FIRST NAME* : Nela

DATE : 27.7.2025

SIGNATURE :



Introduction

Plastics are man-made polymeric materials derived from fossil fuels. Their low cost, durability, flexibility and lightweight properties made them rapidly widespread in the 1970s. Today, plastic materials are used in everyday life by the majority of the population. Although the lifespan of plastics is estimated to be hundreds of years, they can degrade slowly due to biological or abiotic processes such as UV irradiation, heat, friction, etc. (Zhao, Zhang, et al. (2022)) eventually breaking down to micro- and even nanometric sizes. The fragmentation of plastics is a significant source of microplastics (MP) in the environment.

Microplastics are particles of plastic with sizes ranging from 1 μm to 5 mm (Frias et al. (2019)). They enter the environment not only from degradation processes but also as primary materials used in cosmetics and paint (Thompson, Courteney-Jones, et al. (2024)). Since 2004, when the term “microplastics” was first introduced (Thompson, Olsen, et al. (2004)), the presence of microplastics has become a major area of study, both due to growing awareness of the prevalence of microplastics in the environment and because increasing plastic production has not been accompanied by an increase in waste-management capacities. Microplastics now raise public health concerns as they have been detected in human and animal digestive, cardiovascular, respiratory, and reproductive systems (review by Roslan et al. (2024)). Inside the body, microplastics release toxic chemicals and heighten the risk for neurodevelopmental disorders, oxidative stress, infertility, and cancer (Lee et al. (2023)). A better understanding of MP behavior in the environment is necessary to assess global regulations to reduce plastic and microplastic pollution.

Microplastics pose a significant threat to ecosystems as they have been found in diverse environments all over the world, including the deepest ocean trenches (Peng et al. (2018)) and remote glaciers (Stefánsson et al. (2021)). MP have been detected at 2877 m altitude in isolated area of the Pyrenees (Allen et al. (2021)), demonstrating that the atmosphere plays a role in microplastic transport. The sources of atmospheric microplastics are poorly characterized, but they are thought to include sea spray, landfills, agricultural practices, urban waste water, road emissions from tire wear and litter, and various residential emissions such as detrition of synthetic clothing (Thompson, Olsen, et al. (2004)). Global atmospheric microplastic transport simulations show considerable variability, by multiple orders of magnitude, with particular disagreement on ocean emissions which have been estimated to range from 1 $Gg/year$ (Yang et al. (2024)) to 10^4 $Gg/year$ (Brahney et al. (2021)).

This uncertainty is due in part to inconsistencies in experimental methods. Over the past decade, numerous sampling and analysis protocols have been developed to identify and characterize microplastics. Many of these are microscopy-based, involving contrast-based digital or spectral image analysis. These techniques count individual microplastic particles and measure their width and length. The concentrations of microplastics reported from microscopy techniques are often expressed as number concentrations i.e. the count of MP particles per volume of the sample matrix. Alternative techniques (e.g. pyrolysis-gas chromatography-mass spectrometry) provide a direct measurement of polymer mass concentration but do not provide MP counts or dimensions. Comparing MP number or mass concentrations across different studies is challenging because, due to experimental limits of detection, different studies target different MP size ranges and shapes within the formal 1 to 5000 μm range. Thus it is crucial to standardize MP concentrations for proper intercomparison and synthesis of results.

This internship project stems from the ongoing collaboration between IGE Grenoble and GET Toulouse to develop a public database of all size-resolved MP observations in the literature that will facilitate intercomparisons. In this report, we illustrate how the particle size distribution measured by individual studies can be used to extrapolate over the whole MP range from 1 to 5000 μm , or any other size range. We have developed two methods for concentration standardization and discuss their limitations and applications. Additionally, we extrapolate concentrations of various observational studies and we emphasize the importance of sampling parameters, which are often incompletely communicated in articles.

Scientific context

Microplastic abundance has been shown to follow a power law distribution, with smaller particles much more prevalent in the environment due to the fragmentation of larger particles (Kooi and Koelmans (2019)). However, it is currently not possible to sample and measure microplastics across the entire size range of 1 to 5000 micrometers using a single experimental technique. A technique’s limits of detection (LODs) dictate the particle sizes, typically defined by the longest dimension, that can be measured. Various experimental techniques have different LODs, which makes results inconsistent from one study to another. This is because studies report MP concentrations only within their specific experimental LODs, which differ from the LODs of other studies and furthermore do not represent the full 1-5000 μm size range. This leads to misleading information about MP concentrations.

One method to obtain MP concentration over the full 1-5000 size range (or any other desired size range) is to extrapolate the observations, assuming that the power-law distribution holds over the entire size range of interest. Since the power-law distribution transforms into a linear function in x,y log-log scale, it seems feasible to determine the parameters of this log-size, log-counts linear distribution.

Even so, studies often group MP count observations into bins and do not provide raw data. Therefore, the information provided by authors is the MP number concentration within the LODs and MP approximate sizes within the bins. Figure 1 illustrates a typical example of a reported MP size distribution from Song et al. (2018). The observed MP dimensions do not begin at a particle size of 1 μm because of the experimental limits of detection. The bin widths are irregular, with wider bins for larger particles as there are fewer large particles in the environment. In this example, bin width increases with increasing particle size from 50 μm to 1000 μm and eventually to 3000 μm . Notably, the narrowest bins sometimes contain fewer particles than expected, as seen in the first three bins of the MP distribution in Figure 1. This may be a result from the limits of detection of the counting technique used, which may have overlooked some particles due to their small size (Kooi and Koelmans (2019)).

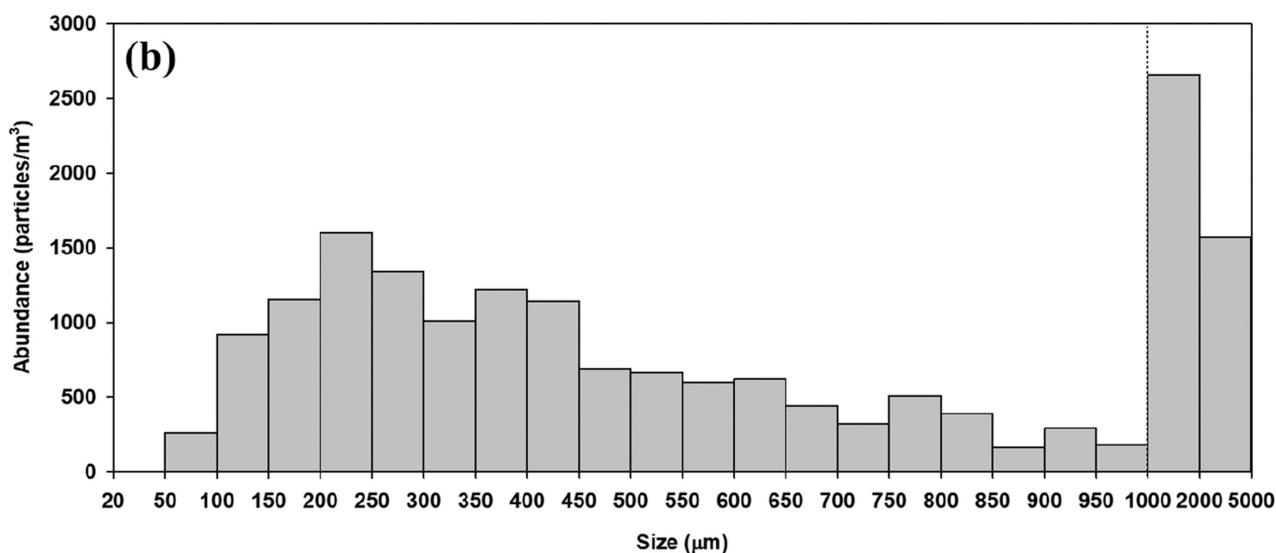


Figure 1: Example of MP size distribution information given by authors in form of a histogram (Song et al. (2018), <https://doi.org/10.1021/acs.est.8b04032>)

My internship focused on evaluating the best way to perform the linear regression fit on log-transformed binned power-law distributions and on estimating the resulting uncertainty. Methods were tested on simulated data and on real observational datasets.

Methods and materials

Data and code availability

All numerical work was done in Python in Jupyter Lab online interface and is accessible via the following link: <https://github.com/IGE-Microplastics/Nela-Dobiasova-2025-internship-results>

Bin normalization method

As mentioned, many studies use highly variable bin widths when reporting MP sizes. This adds a difficulty to comparing concentrations between studies (Leusch et al. (2023)). To make sure the size of the bins doesn't affect the particle counts, Leusch et al. (2023) proposed to divide the number of particles in each bin by the bin width. This way, we can account for the fact that a smaller number of particles is contained in bins with increasing width. In this report, we built on their work to quantify the limits of this "bin normalization" method.

Bin number	Lower limit (μm)	Upper limit (μm)	Midpoint (μm)	MP counts	Normalized MP counts (1/μm)	Cumulative MP counts
1	200	250	225	1500	30	9050
2	250	300	275	1300	26	7550
3	300	350	325	1000	20	6250
4	350	400	375	1250	25	5250
5	500	1000	750	2500	5	4000
6	1000	2000	1500	1500	1.5	1500

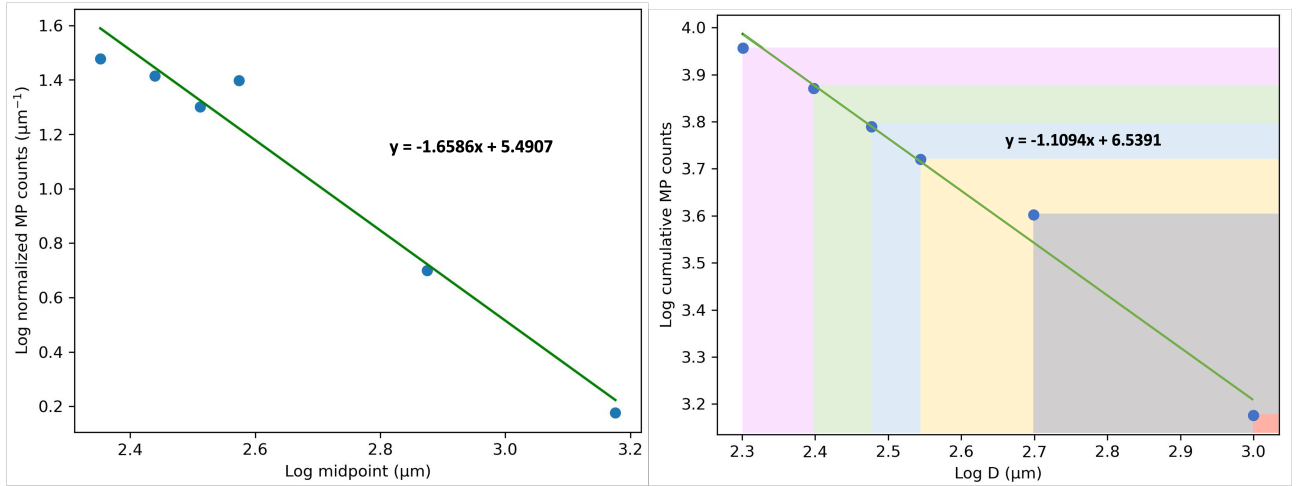


Figure 2: Illustrative example inspired by published studies with Table of MP counts (top), Normalized size distribution in x,y logarithmic scale with corresponding linear equation (left bottom), and Cumulative sum of particles larger than threshold D with sizes taken in account represented by colored rectangles and associated linear equation (right bottom); green line represents the linear regression

Figure 2 (top) represents an illustrative example of the calculations of normalizing MP counts by bin width. We consider that all particles inside the bin have the size of the bin's midpoint. Plotting normalized bin counts as a function of the bin midpoint x in a log-log scale would give us a linear function (Eq 1, Figure 2 left bottom) which can be converted to the power-law function (Eq 2). We can then extrapolate the observed counts (n) over the full 1-5000 μm range (N) using Eq 3.

$$\log(count) = a \cdot \log(x) + b \quad (1)$$

$$count = 10^b \cdot x^a \quad (2)$$

$$N_{1-5000} = \int_1^{5000} 10^b \cdot x^a \, dx \quad (3)$$

Cumulative sum method

Since we are mainly interested in determining the total number of particles in the 1-5000 μm range, using a cumulative density function is an intuitive approach. Clauset et al. (2009) detail the application of cumulative sum on power-law distributions and show its great efficiency. Here, based on their analysis, we use this method to extrapolate MP concentrations from observational datasets.

The cumulative sum method focuses exclusively on the information provided by authors: the number of measured particles in each bin and the number of observed particles n . We replace the MP count of each bin with $N_{>D}$, the cumulative count of all particles larger than the bin's lower limit D (see Figure 2 last column and Figure 2 right bottom). This method results also in a linear function in a log-log scale (Eq 4), allowing the determination of N_{1-5000} , the number of particles in the full 1 to 5000 μm size range using equations 5 and 6:

$$\log(count_{cum}) = a_{cum} \cdot \log(D) + b_{cum} \quad (4)$$

$$count_{cum} = 10^{b_{cum}} \cdot D^{a_{cum}} \quad (5)$$

$$\begin{aligned} \log(N_{>1}) &= a_{cum} \cdot \log(1) + b_{cum} \\ \log(N_{>5000}) &= a_{cum} \cdot \log(5000) + b_{cum} \\ \rightarrow N_{1-5000} &= 10^{\log(N_{>1})} - 10^{\log(N_{>5000})} \end{aligned} \quad (6)$$

The link between the slopes a and a_{cum} can be determined thanks to the statistical proprieties of probability density function (PDF) and the complementary cumulative density function (CDF):

PDF :

$$\begin{aligned} p(x) &= 10^b \cdot x^a \\ \rightarrow \log(counts) &= a \cdot \log(x) + b \end{aligned}$$

CDF :

$$\begin{aligned} P(x) &= \int_x^\infty p(x) \, dx \\ &\propto x^{(a+1)} \\ \rightarrow \log(counts) &\propto (a+1) \cdot \log(x) \end{aligned}$$

The intercept b can be found using the result of the cumulative sum

$$\begin{aligned} N_{1-5000} &= \int_1^{5000} 10^b \cdot x^a \, dx \\ \rightarrow b &= \log_{10}\left(\frac{N_{1-5000} \cdot (a+1)}{5000^{a+1} - 1^{a+1}}\right) \end{aligned}$$

For more details and further explications on probability density functions of power law distributions on binned data see (Virkar et al. (2014)).

Simulated sampling model

To evaluate both methods, a code was developed to simulate microplastics sampling out of an environmental matrix, assuming the MP sizes follow a perfect power law distribution $MP\ count = 10^b \cdot size^a$. The input parameters for the simulation are 1) the slope of the distribution (a); 2) the lower and upper limits of detection (LODs); and 3) the number of sampled MP (n). In addition, one has to define the bins in which sampled particles are stored.

First, the particle distribution within the matrix was determined using `np.random.uniform` function which produced n random values between 0 and 1. Next, a particle size within the LODs range was associated with these random values by applying the power law function based on the input slope. Then, all particles were selected and divided into the predetermined bins. The counts of MPs in each bin constitute the model's output. It's important to note that for fixed input parameters, each call of the `np.random.uniform` function initiates a new "sampling campaign."

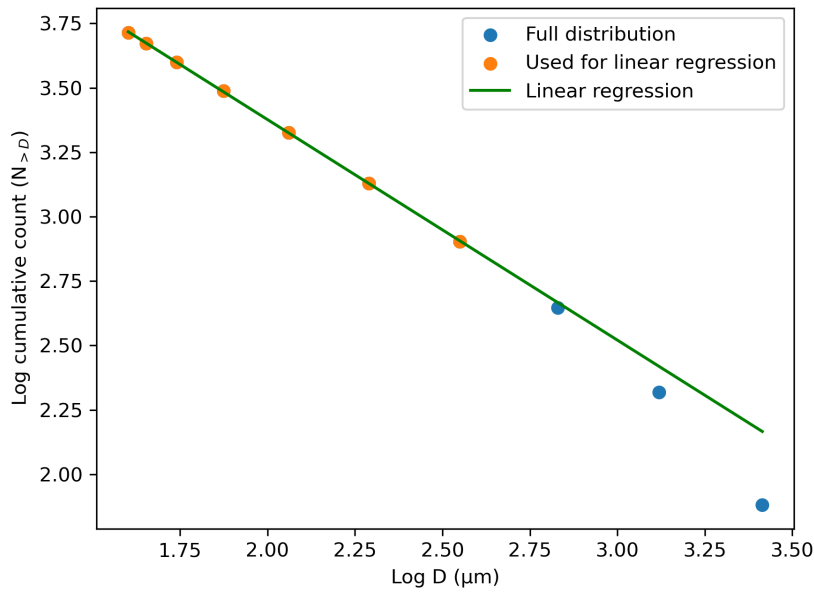


Figure 3: Cumulative counts and their corresponding threshold values D plotted in x,y logarithmic scale (blue) with linear regression (green) and values on which the linear regression was calculated (orange)

After that, the chosen method (either bin normalization or cumulative sum) is applied and linear regression is plotted. Given that there is only a limited number of particles at the tail of the distribution, the last bins usually don't follow the expected linear function as illustrated in Figure 3 where points of the distribution not following linear function are showed in blue. The tail bins contain fewer particles compared to other bins and the relationship between counts and bin width changes such that the big particles look under-represented. Consequently, we visually choose the bins that correspond to the linear portion of the log-log distribution, using only these bins to determine the slope (orange points in Figure 3).

Finally, the outputted slope is compared with the initial input value. The error in the estimated slope is calculated as the absolute difference between the estimated slope and the true slope. Note that, the input parameter is n (the number of sampled particles) instead of N (the total number of particles in the full 1-5000 μm size range). This is done to ensure better control over the number of sampled particles. Thus, the output N is compared to the average value of outputted N across all sampling campaigns, and the error associated with N is determined using Equation 7. The error on N is expressed as a mean absolute fraction error.

$$N_{error} = \frac{\text{mean} | N_{output} - N_{mean} |}{N_{mean}} \text{ where } N_{output} \text{ are the output values for each sampling campaign} \quad (7)$$

Counts to mass conversion

Microplastics are highly studied particles in various scientific fields. The number of microplastics is an important quantification for fields, such as toxicology, interested in MP specific surface area likely to react with other bodies. Whereas, for atmospheric or medical studies, a significant measure of MP pollution is the total microplastics mass. Thus, the conversion from counts to mass is an necessary step to characterize the method’s efficiency.

Once the slope of the power law distribution and total microplastics count (N) are established, the associated mass can be estimated. Given the value of N and the slope, a particle size distribution is generated using the simulated sampling model. This size distribution is then converted into a mass distribution, and the total mass in a matrix is calculated as the average mass from 200 sampling campaigns.

Since studies often provide only one dimension of MP (major axis, spherical equivalent diameter), the conversion from MP counts to MP mass relies on morphological studies (Koelmans et al. (2020), Kooi, Primpke, et al. (2021)). They typically observe the following dimension ratios: $width = 0.7 \cdot length$ and $height = 0.7 \cdot width$ which are used in MP volume estimation in this report. For simplification—and due to the frequent lack of precise information—MP are assumed to have an ellipsoidal shape. To estimate the mass of microplastics, the density is assumed to be 1 g/cm^3 which is commonly associated with polymers.

Results and discussion

Bin normalization method

When analyzing the bin normalization method, it was discovered that the slope depends heavily on the chosen bin widths. In fact, the slope value is intrinsically linked to the binning selected by the authors. As there is no standardized approach to bin particles, the bin widths can vary significantly from one study to another. For instance, in the illustrative example (Figure 2), if bins 1 and 2 and bins 3 and 4 were combined, creating two bins of 100 μm each instead of four bins of 50 μm , the slope would decrease by 0.045 passing from -1.659 to -1.703. This change would correspond to a difference of 107,000 particles across the full 1-5000 μm size range. Moreover, the steeper the slope, the larger the potential error in estimated N. Figure 4 illustrates how both slope (left) and N (right) are influenced by the bins used on observed MP size data, highlighting the uncertainty in the value of N. It shows that depending on selected bin width, N can vary by multiple orders of magnitude. Therefore, a more in-depth exploration of the relationship between slope values and bin widths—considering varying input parameters such as slope, LODs, and N—was necessary.

The concept of optimal bin width – defined as a bin width that minimizes the error in slope – was developed in order to identify an efficient binning method for future studies as well as for rebinning existing data. Firstly, the optimal bin width differs with varying input parameters. For instance, a simulation with input parameters LODs, slope, and n, gives a different result when any one of the three parameters changes. This behavior is illustrated in Figure 5 where the left side shows an optimal bin width based on LODs ranging from 50 to 4500 μm , while the right side presents the same slope and sample size but with LODs 10 to 150 μm . The optimal bin width for the first case scenario would be 280 μm (intersection of mean output slope and the input slope) and for the second case, with smaller particles sampled, it would be around 10 μm . Even more, the optimal bin width is specific to each sampling campaign. The shaded regions in the figures represent the range of optimal bin widths across 200 sampling campaigns, indicating that the ideal bin width falls between 90 μm and 550 μm

for the left graph, and between 5 μm and 30 μm for the right graph. A bin width of 50 μm in the right graph corresponds to only 2 bins used for slope determination.

As a result of our analysis, no universal optimal bin width minimizing the error on slope was found, leading to the rejection of the bin normalization method due to its substantial and difficult-to-estimate error in extrapolated MP concentration.

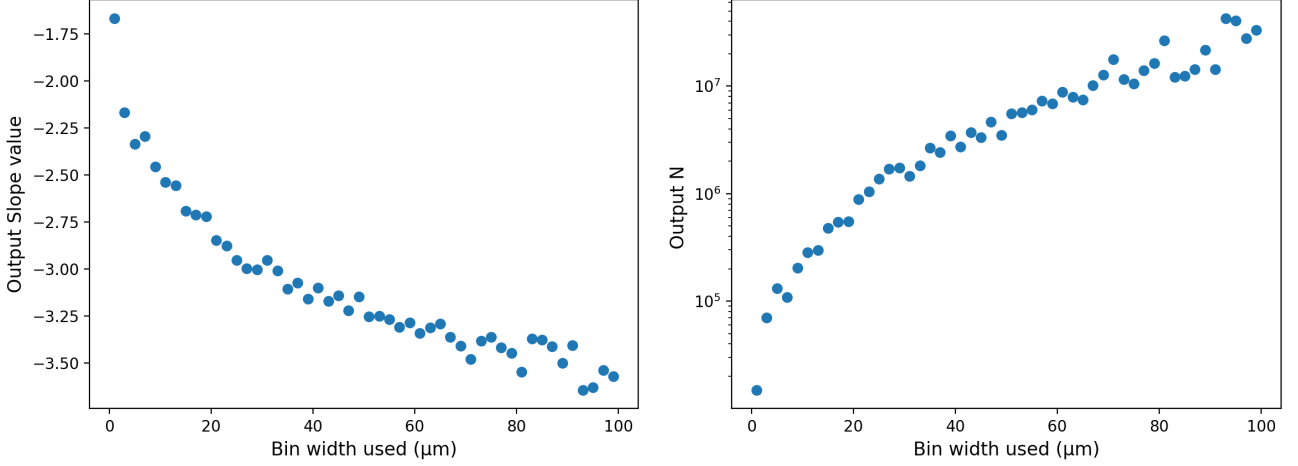


Figure 4: Dependence of slope (left) and of the total number of particles N in the full 1-5000 μm size range (right) on chosen bin width. Particle size distribution from raw data containing MP counts and dimensions provided by Ueda et al. 2025 (site D, fragments, pump method only), <https://doi.org/10.1016/j.rsma.2025.104108>

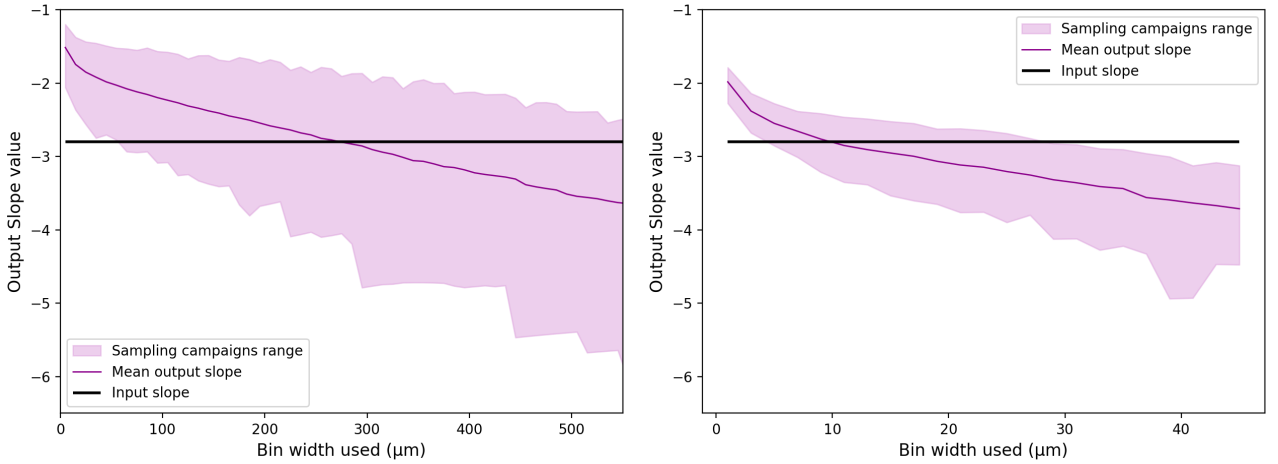


Figure 5: Influence of bin width on the difference between the input and estimated (output) slope of power-law-distributed particle sizes with input parameters: slope = -2.2, $n=800$, LODs = [50,4500] (left), LODs = [10,150] (right); both using bin normalization method. Shaded regions show the range across 200 sampling campaigns, with mean value shown in dark violet color.

Cumulative sum method

The cumulative sum method, when applied with different bin widths, demonstrates less variability in slope than the bin normalization method. This is seen in Figure 6, which uses the same input parameters as Figure 5 left, but employs the cumulative sum method. The cumulative sum method has consistently low error in slope, thus reducing the error in the MP concentration N compared to the bin normalization method.

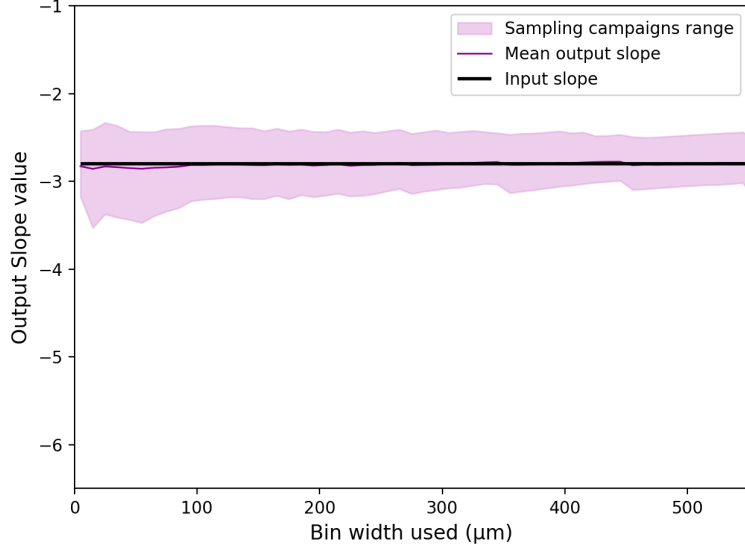


Figure 6: Influence of bin width on the difference between the input and estimated (output) slope of power-law-distributed particle sizes with input parameters : slope = -2.2, $n=800$, LODs = [50,4500]; using cumulative sum method. Shaded regions show the range across 200 sampling campaigns, with mean value shown in dark violet color.

Although the error in slope, and consequently N , of the cumulative sum method is always lower than the error of the bin normalization method, it is still influenced by the input parameters. A deeper analysis of the variability of slope error leads to a 3D matrix (Figure 7) that estimates slope errors based on input parameters. In Figure 7 top, the x, y, and z axes represent input slope, input lower limit of detection, and input number of sampled particles - n , respectively, with the upper LOD set to 5000. The values represented by the color map correspond to the mean value of absolute errors on slope over 200 sampling campaigns using the simulated sampling model.

Figure 7 middle left provides a 2D transect of this 3D field, representing the horizontal plane intersecting the n -axis at 100 particles. For instance, with a shallow slope (slope = -1.1), the error in slope increases when only larger particles are considered (LOD=227 μm). However, if enough particles are sampled and the power law is strongly pronounced (slope < -2), the error remains relatively low even if the LOD increases to 505 μm (as shown in Figure 7 middle right).

In summary, slope steepness has the greatest impact on error; regardless of the number of particles sampled or the limits of detection. The estimated slope error is larger if the power law distribution is less pronounced (slope closer to -1). The second most significant factor affecting slope error is the lower LOD. Logically, when only larger particles are sampled, it becomes challenging to accurately estimate the distribution of the smaller particles which dominate the counts.

The information given by our simulated samplings can be used to estimate the error in slope values. For instance, for an imaginary study that observed 2000 particles in the range between 505 and 5000 μm , we can estimate that the error on the given slope would be between 0.06 and 0.5 as illustrated in Figure 7 (bottom) showing the error values for fixed n and lower LOD.

The error associated with the slope is not directly proportional to the error in N . In fact, a steeper slope (with a value of -3.4) results in a greater error in N compared to a shallower slope (with a value of -1.1). Even though a shallower slope results in a larger slope error, it doesn't necessarily imply that there is a significant error in N . The mean absolute fraction error on N (calculated using Eq 7) is illustrated on Figure 8, where orange and red colors indicate that the mean absolute difference between the estimated and the "true" value was greater than the true value. Figure 8 bottom includes only those input combinations that lead to an error greater than the "true" value, showing which input parameter combinations do not provide sufficiently precise information. In fact, these combinations often provide an output N significantly greater than the mean N .

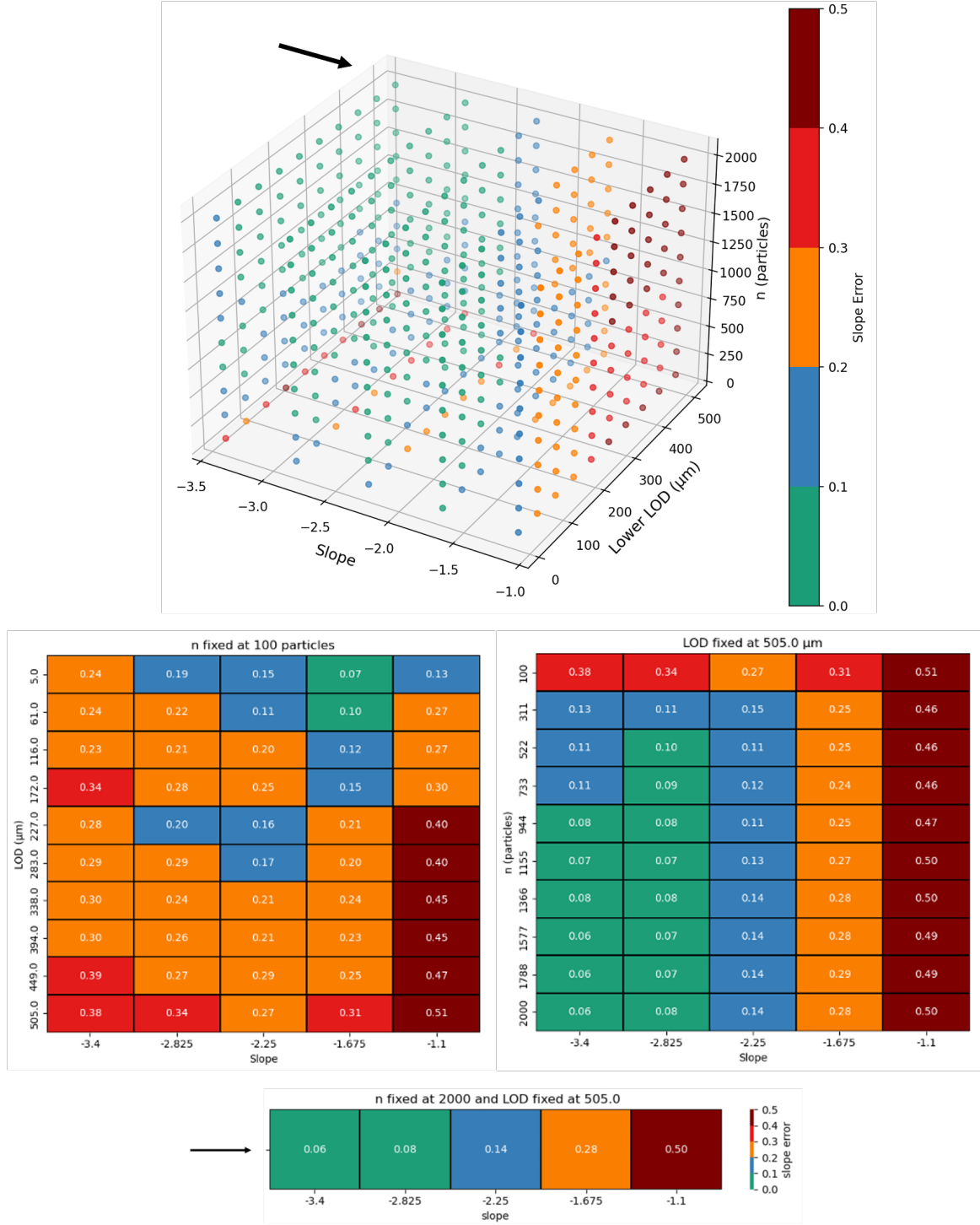


Figure 7: Variability of slope error with changing model's input parameters; slope, sample size n and lower limit of detection. Upper LOD is set to 5000 μm . 3D representation of slope error values represented by color map (top), 2D horizontal plane intersecting z axis at 100 particles (middle left), 2D vertical plane intersecting y axis at 505 μm (middle right), and 1D intersection of $n = 2000$ and $\text{LOD} = 505 \mu\text{m}$ planes indicated by black arrow in the 3D field (bottom)

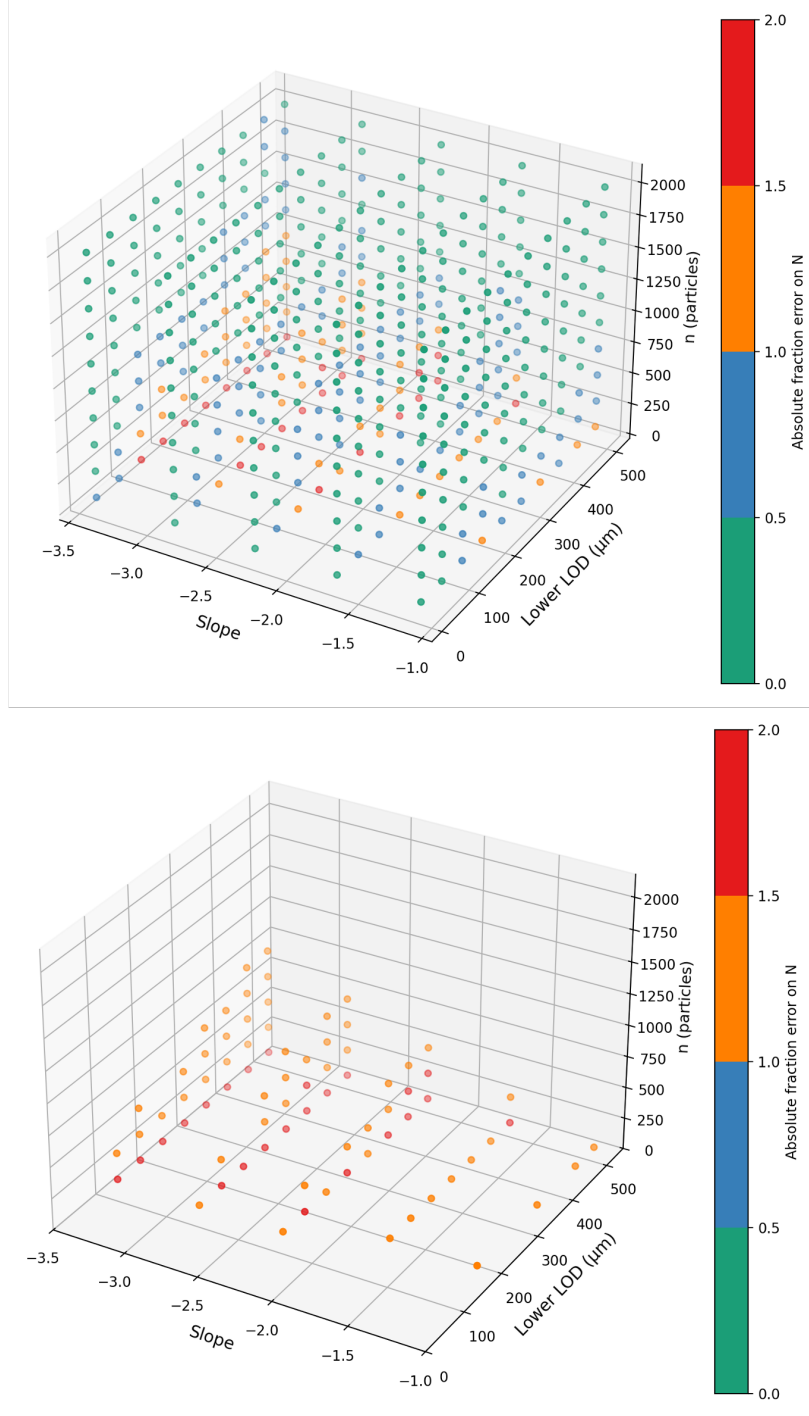


Figure 8: Variability of mean absolute fraction error on N (top) with changing model's input parameters; slope, sample size n and lower limit of detection. Only values superior to 1 are visualized (bottom). Upper LOD is set to 5000 μm .

An important limitation of the cumulative sum method is the necessity for a visual selection of bins to perform linear regression as previously noted by Virkar et al. (2014). Large particles are less frequent resulting in fewer particles in the tail bins compared to their width. In observational datasets, even smaller particles can be underestimated due to the challenges associated with detecting them, particularly in visual measurement techniques. Including tail bins tends to steepen the slope, leading to a systematic overestimation of particle numbers (N). Conversely, including the bins at the beginning of the distribution with underestimated counts tends to lower the slope and result in an underestimation of N .

While automated threshold determination is possible for simulated samplings, real observational data are more complex and require human judgment. Figure 9 presents two examples of binned counts from Dris et al. (2016) (left) and Gaston et al. (2020) (right), highlighting the challenges in accurately identifying the linear segment of the log-log size distribution in observational datasets. The subjective selection of this linear segment introduces an error that is difficult to quantify.

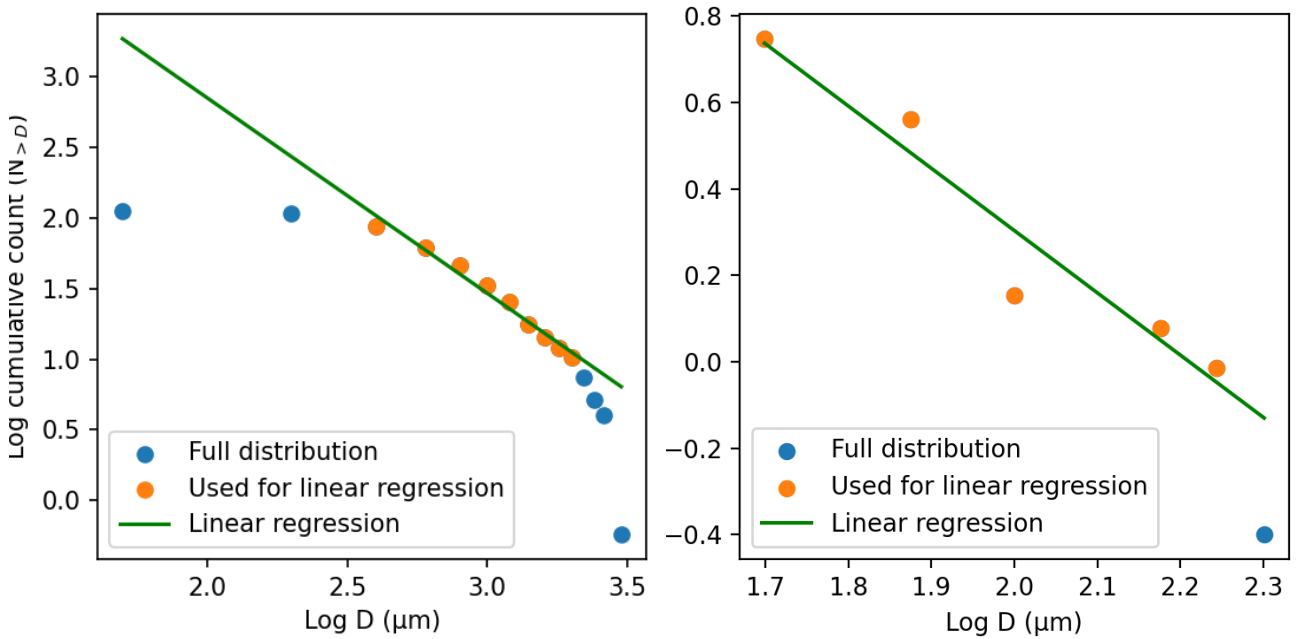


Figure 9: Cumulative sum method applied on real data from Dris et al. (2016) (left) and Gaston et al. (2020) (right) – x,y log log representation of threshold D and associated cumulative counts

Overall, the cumulative sum method, as presented in this report, has been proven to be effective for intercomparisons and provides sufficient error estimations. Nevertheless, further refinements could be made to reduce even more the errors. For example, the total number of particles across the interval of interest is calculated by extrapolating the cumulative sum distribution (Eq 6). This value is then used to determine the “b” parameter of the $10^b \cdot x^a$ distribution. An adjustment could be made by imposing $N^{lower\ LOD} - N^{upper\ LOD}$ to be equal to n – the number of sampled particles reported by the study. Imposing this constraint might allow the distribution to be better aligned with the observed distribution and reduce errors in N . This hypothesis requires further exploration.

However, due to the lack of precise information regarding limits of detection, the lower limit of the first bin does not necessarily represent the smallest particle detected, and thus it may not be the true limit of detection. Similarly, the upper limit of the last bin does not necessarily represent the largest particle found. These details regarding limits of detection can significantly affect error estimations and pose a considerable constraint on proposed method adjustments.

Application of cumulative sum method on real data

First, the cumulative sum method was applied to real data provided by Ueda et al. (2025), which includes the sizes of individual microplastic particles collected from surface water samples in four sites in Tokyo Bay, Japan. Two sampling methods were employed, but this report focuses on samples collected using the hand pump method because of its smaller lower limit of detection.

Particles were binned using various binning methods, primarily irregular bins, starting with small bin widths and increasing for larger particles. The results showed similar concentrations across all binning strategies with mean absolute fraction error on N , due to binning, of approximately 0.18 (mean over the four similar sites). The estimated error attributed to sampling parameters (number of sampled particles and LODs) reached up to 0.56 (maximum of the 4 sites). This low error can be attributed to the lower LOD, which is 21 μm for all sites.

Table 1 summarizes the total number of particles for each site. N (MP/m³) is a corrected value according to the water volume that has been analyzed. The variations in particle counts between sites may be influenced by their distance from the coast, with site C having the largest distance and site A the smallest. Notably, site D is located near or beneath a heavily used car bridge, which may be a significant source of microplastics in that area. The majority of microplastics found at site D were made of poly(methyl methacrylate), a material commonly used in road marking paints and car coatings, lending support to this hypothesis.

Site	A	B	C	D
number of MP measured	610	484	139	2219
lower LOD	21	21	21	21
upper LOD	502	1495	562	658
N (MP/m ³)	$12.7 \cdot 10^6$	$8.0 \cdot 10^6$	$8.0 \cdot 10^6$	$20 \cdot 10^6$
mean absolute fraction on N	0.33	0.35	0.56	0.17

Table 1: Sampling parameters of 4 sites in Tokyo Bay analysed by Ueda et al. (2025) and calculated microplastic concentration, using cumulative sum method. N (MP/m³) stays for MP abundance in the full range, taking in account the amount of water analyzed and represents the concentration in MP per m³ of water.

Second, the cumulative sum method was applied to various observational studies that published binned data. Figures 10 and 11 clearly illustrate the significance of extrapolating measured concentrations as they compare three different studies that report similar microplastics (MP) concentrations.

Tekman et al. (2020) focused on Arctic Sea surface, while Zhu et al. (2021) conducted sampling campaigns in North Pacific Subtropical Gyre (this report presents only subsurface concentrations from depth of 3400 m). Zhao, Mincer, et al. (2023) studied airborne MP concentrations (this report presents only concentrations measured in Shanghai). All three observations report concentrations around 265 MP per m³ of sampled matrix, but the MPs were collected across different LOD ranges.

In Figures 10 and 11, the sampling range of each study, defined by LODs (detailed in *LODs reported* column in the tables), is indicated by the width of the colored horizontal lines. The vertical axis displays the extrapolated MP concentrations calculated using the cumulative sum method (N_{1-5000} or M_{1-5000} shown in the tables). Shaded regions represent the estimation errors, calculated using the mean absolute fraction error (N error or M error in the tables). In Figure 10, the left graph shows that the extrapolated MP concentration from Zhu et al. (2021) is significantly higher than those from the other two studies, even though all three studies report similar measured concentrations. This notable difference arises partly from the LOD but primarily from the steepness of the slope. Due to its steeper slope, the extrapolated MP concentration from Zhu et al. (2021) results in a high number of small particles. While small particles dominate numerically, they contribute minimally to total mass. The right graph of Figure 10 reveals that total mass concentration is similar across the three studies, with Zhu et al. (2021) study accounting for the smallest mass. A few large particles from both Zhu et al. (2021) and Zhao, Mincer, et al. (2023) studies drive the majority of the mass concentration.

However, different research fields are interested in different MP sizes. Environmental and geo-

ecological studies often focus on larger particles due to their mass, which facilitates sedimentation and benthic accumulation (Soler et al. (2025)). In contrast, medical and toxicological research is more concerned with smaller microplastics and nanoplastics. To conclude on efficiency of our model on different size ranges, an extrapolation over 1-100 μm size range was equally analyzed.

The concentration of small microplastics is illustrated in Figure 11. A comparison of Figures 11 and 10 shows that most particles have lengths inferior to 100 μm . Due to their small size, these MP represent around one-thousandth of the total mass for both Tekman et al. (2020) and Zhao, Mincer, et al. (2023)). The quantity of MP plays an important role in mass comparisons, explaining the predominance of the mass concentration observed in the Zhu et al. (2021) study (see Figure 11).

Note that error made while selecting bins for linear regression was not yet quantified and may be of a great influence on presented results. Plotted linear regressions are available in Supplementary information section.

Author	MP/ m^3 reported	LODs re- ported	LODs used	N_{1-5000} (MP/ m^3)	N error	M_{1-5000} ($\mu\text{g}/\text{m}^3$)	M error
Tekman (2020)	239	10–100	10–100	2749	0.25	1752	1.72
Zhao (2021)	297	25–500	50–500	1648830	0.77	562	1.51
Zhu (2023)	260	5–1000	5–1000	3427	0.16	1064	1.56

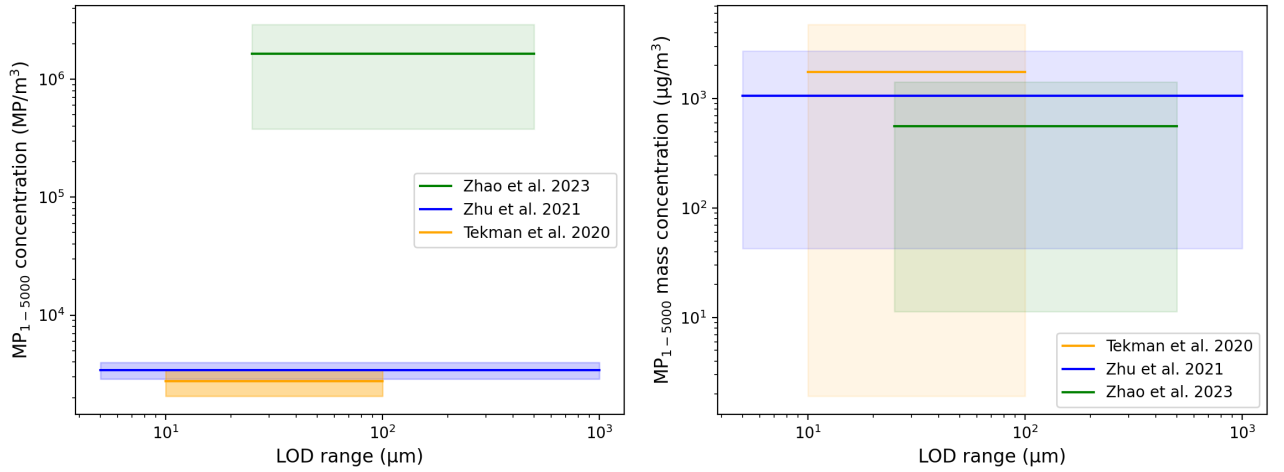


Figure 10: Comparison of observed MP number (N) and mass (M) concentrations extrapolated over the **1-5000 μm** size range of three different observational studies: Tekman et al. (2020), Zhu et al. (2021) - water depth 3400 m, Zhao, Mincer, et al. (2023) - site Shanghai, reporting similar measured concentrations (MP/ m^3 reported). X axis represents MP sizes observed by each study (LODs reported). Y axis shows estimated N_{1-5000} (left) and estimated Mass_{1-5000} (right) using cumulative sum method. Shaded regions correspond to estimated error range (N error and M error corresponding to mean absolute fraction errors).

Author	MP/m ³ reported	LODs re- ported	LODs used	N ₁₋₁₀₀ (MP/m ³)	N error	M ₁₋₁₀₀ (μg/m ³)	M error
Tekman (2020)	239	10–100	10–100	2736	0.23	1.44	0.41
Zhao (2021)	297	25–500	50–500	552898	0.88	16.95	0.56
Zhu (2023)	260	5–1000	5–1000	3417	0.17	1.29	0.37

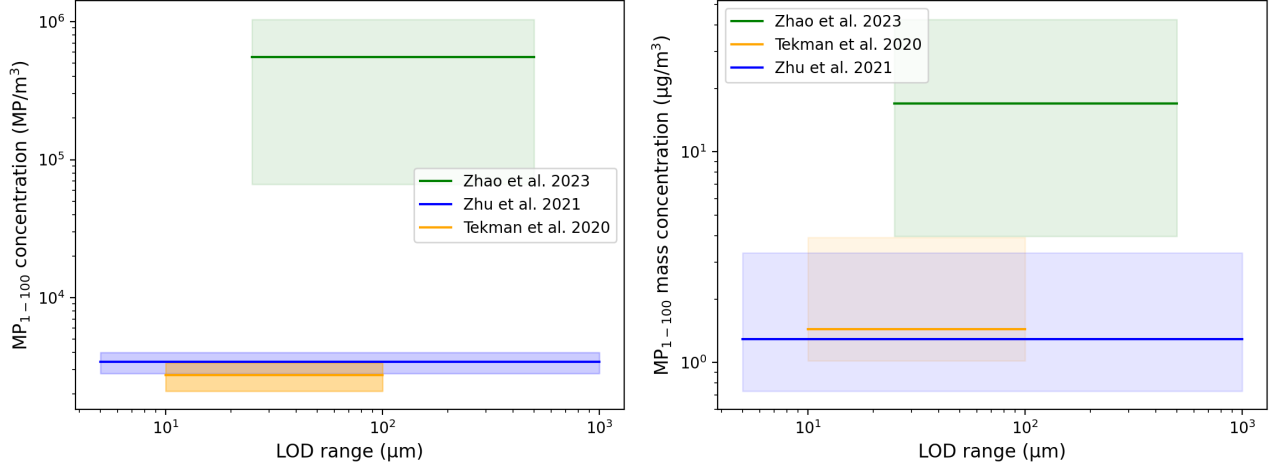


Figure 11: Comparison of observed MP number (N) and mass (M) concentrations extrapolated over the **1-100 μm** size range of three different observational studies: Tekman et al. (2020), Zhu et al. (2021) - water depth 3400 m, Zhao, Mincer, et al. (2023) - site Shanghai, reporting similar measured concentrations (MP/m³ reported). X axis represents MP sizes observed by each study (LODs reported). Y axis shows estimated N₁₋₁₀₀ (left) and estimated Mass₁₋₁₀₀ (right) using cumulative sum method. Shaded regions correspond to estimated error range (N error and M error corresponding to mean absolute fraction errors).

Conclusion

Our work has expanded the observations made by Leusch et al. (2023), who introduced the bin normalization method on observational binned MP counts. Their research demonstrated that normalizing counts by bin width reduces error in the slope compared to not normalizing at all. However, a significant uncertainty concerning the resulting slope remains. We have observed that even a small uncertainty in slope can lead to a substantial error in the estimated number of particles, changing its value by several orders of magnitude (Figure 4).

We tested this method using simulated datasets binned in various bin widths. These “simulated samplings” allowed us to compare the resulting slope with the input, real, slope. Our findings indicated that the error in the estimated slope depends heavily on the chosen bin widths (Figure 5). Accurately estimating this error is challenging without prior knowledge of the true slope. Consequently, we found the bin normalization method to be inadequate.

The cumulative sum method demonstrated a smaller dependence on bins used and lower error in the final results. Since the cumulative sum’s accuracy is not affected by the chosen bin widths, we were able to estimate the error in the output slope and output N (Figure 7 and Figure 8). This error is highly influenced by the slope steepness and lower limit of detection. It is evident that datasets reporting only a few large microplastics will result in a high uncertainty regarding the number of particles in the full 1-5000 μm size range, which is mostly comprised of smaller particles. This is because determining the slope of a power-law distribution based on a few particles in the tail is quite difficult.

The information regarding N and slope error derived from our simulated samplings can help evaluate the reliability of calculated MP abundance. Experimental studies with higher LOD that

observe only larger particles are not necessarily futile; the range of interest is not always the full 1 to 5000 μm range as it varies based on research objectives. Therefore, the error estimation based on simulated samplings can be of great help in selecting studies that provide sufficient information for accurate extrapolation over the size range of interest.

The application of the cumulative sum method reveals important insights into microplastic concentrations across different studies, despite similar reported values, and highlights once again the importance of extrapolating observed MP concentrations (Figure 10, Figure 11). Our analysis on three observational datasets emphasizes the need to consider both the size distribution of microplastics and the limits of detection when interpreting results. Notably, while smaller particles may dominate numerically, they contribute less to total mass compared to larger particles.

Finally, the primary challenge in accurately estimating microplastic abundance often stems from the limited information shared by researchers. To enhance the precision of these estimations, it's crucial for studies to present more comprehensive MP parameters. Many studies currently show small counts across a small number of bins. Developing a consensus on effective communication of observed microplastics, could significantly improve the extrapolation precision. Ideally, researchers should consider sharing raw, unbinned data to facilitate better comparisons which are essential for advancing our knowledge on microplastic pollution.

Acknowledgments

I would first like to express my gratitude to my internship supervisor, Ian Hough, for his time, encouragement, and guidance throughout this internship. My thanks also go to Jeroen Sonke and Theo Segur in GET Toulouse, who gave me all the support I needed. It is thanks to their MP database that I could work on this topic. Thank you for all your advice and insights on research work. Working with these three was an incredible experience. I should not forget to thank my colleague Didier Voisin for his bright ideas. Finally, I would also like to thank the entire CHIANTI team for their warm welcome at IGE, especially Jennie Thomas and Hélène Angot. Thank you for letting me be part of this microplastics journey.

References

- Allen, S., D. Allen, F. Baladima, V.R. Phoenix, J.L Thomas, G. Le Roux, and J.E. Sonke (2021). “Evidence of free tropospheric and long-range transport of microplastic at Pic du Midi Observatory”. In: *Nat Commun.* 12.1. DOI: [10.1038/s41467-021-27454-7](https://doi.org/10.1038/s41467-021-27454-7).
- Brahney, J., N. Mahowald, M. Prank, G. Cornwell, Z. Klimont, H. Matsui, and K. A. Prather (2021). “Constraining the atmospheric limb of the plastic cycle”. In: *Proceedings of the National Academy of Sciences of the United States of America* 118.16. DOI: [10.1073/pnas.2020719118](https://doi.org/10.1073/pnas.2020719118).
- Clauset, A., C. R. Shalizi, and M. E. J. Newman (2009). “Power-Law Distributions in Empirical Data”. In: *SIAM Review* 51.4. DOI: [10.1137/070710111](https://doi.org/10.1137/070710111).
- Dris, R., J. Gasperi, C. Mirande, C. Mandin, M. Guerrouache, V. Langlois, and B. Tassin (2016). “A first overview of textile fibers, including microplastics, in indoor and outdoor environments”. In: *Applied Spectroscopy* 74.6. DOI: [10.1016/j.envpol.2016.12.013](https://doi.org/10.1016/j.envpol.2016.12.013).
- Frias, J.P.G.L. and R. Nash (2019). “Microplastics: Finding a consensus on the definition”. In: *Marine pollution bulletin* 138. DOI: [10.1016/j.marpolbul.2018.11.022](https://doi.org/10.1016/j.marpolbul.2018.11.022).
- Gaston, E., M. Woo, C. Steele, S. Sukumaran, and S. Anderson (2020). “Microplastics Differ Between Indoor and Outdoor Air Masses: Insights from Multiple Microscopy Methodologies”. In: *Environmental Pollution* 221. DOI: [10.1177/0003702820920652](https://doi.org/10.1177/0003702820920652).
- Koelmans, A. A., P. E. Redondo-Hasselerharm, N. H. M. Nor, and M. Kooi (2020). “Solving the Nonalignment of Methods and Approaches Used in Microplastic Research to Consistently Characterize Risk”. In: *Environ. Sci. Technol.* 54.19. DOI: [10.1214/13-aos710](https://doi.org/10.1214/13-aos710).
- Kooi, M. and A. A. Koelmans (2019). “Simplifying Microplastic via Continuous Probability Distributions for Size, Shape, and Density”. In: *Environ. Sci. Technol. Lett.* 6.9. DOI: [10.1021/acs.estlett.9b00379](https://doi.org/10.1021/acs.estlett.9b00379).
- Kooi, M., S. Primpke, S. M. Mintenig, C. Lorenz, G. Gerdt, and A. A. Koelmans (2021). “Characterizing the multidimensionality of microplastics across environmental compartments”. In: *Water research* 202. DOI: [10.1016/j.watres.2021.117429](https://doi.org/10.1016/j.watres.2021.117429).
- Lee, Y., J. Cho, J. Sohn, and C. Kim (2023). “Health Effects of Microplastic Exposures: Current Issues and Perspectives in South Korea”. In: *Yonsei Med J* 64.5. DOI: [10.3349/ymj.2023.0048](https://doi.org/10.3349/ymj.2023.0048).
- Leusch, F. DL., H.-C. Lu, K. Perera, P. A. Neale, and S. Ziajahromi (2023). “Analysis of the literature shows a remarkably consistent relationship between size and abundance of microplastics across different environmental matrices”. In: *Environ. Pol.* 319. DOI: [10.1016/j.envpol.2022.120984](https://doi.org/10.1016/j.envpol.2022.120984).
- Peng, X., M. Chen, S. Chen, S. Dasgupta, H. Xu, K. Ta, M. Du, J. Li, Z. Guo, and S. Bai (2018). “Microplastics contaminate the deepest part of the world’s ocean”. In: *Geochemical perspectives letters* 11. DOI: [10.7185/geochemlet.1829](https://doi.org/10.7185/geochemlet.1829).
- Roslan, N. S., Y. Y. Lee, Y. S. Ibrahim, S. T. Anuar, K. M. K. K. Yusof, L. A. Lai, and T. Brentnall (2024). “Detection of microplastics in human tissues and organs: A scoping review”. In: *J Glob health*.
- Soler, M., J. Colomer, F. Pohl, and T. Serra (2025). “Transport and sedimentation of microplastics by turbidity currents: Dependence on suspended sediment concentration and grain size”. In: *Environ. Internat.* 195. DOI: [10.1016/j.envint.2025.109271](https://doi.org/10.1016/j.envint.2025.109271).
- Song, Y. K., S. H. Hong, S. E. Jang, G. M. Han, A. Isobe, and W. J. Shim (2018). “Horizontal and Vertical Distribution of Microplastics in Korean Coastal Waters”. In: *Environ. Sci. Technol.* 52.21. DOI: [10.1021/acs.est.8b04032](https://doi.org/10.1021/acs.est.8b04032).
- Stefánsson, H., M. Peterzell, M. Konrad-Schmolke, H. Hannesdóttir, E. J. Ásbjörnsson, and E. Sturkell (2021). “Microplastics in Glaciers: First Results from the Vatnajökull Ice Cap”. In: *Sustainability* 13.3183. DOI: [10.3390/su13084183](https://doi.org/10.3390/su13084183).
- Tekman, M. B., C. Wekerle, C. Lorenz, S. Primpke, C. Hasemann, G. Gerdt, and M. Bergmann (2020). “Tying up Loose Ends of Microplastic Pollution in the Arctic: Distribution from the Sea Surface through the Water Column to Deep-Sea Sediments at the HAUSGARTEN Observatory”. In: *Environ. Sci. Technol.* 54.7. DOI: [10.1021/acs.est.9b06981](https://doi.org/10.1021/acs.est.9b06981).
- Thompson, R. C., W. Courteney-Jones, J. Boucher, K. DavRaubenheimer, S. J. Rowland, A. W. G. John, D. McGonigle, and A. A. Koelmans (2024). “Twenty years of microplastic pollution research—what have we learned?” In: *Science* 286.6720. DOI: [10.1126/science.adl2746](https://doi.org/10.1126/science.adl2746).

- Thompson, R. C., Y. Olsen, R. P. Mitchell, A. Davis, S. J. Rowland, A. W. G. John, D. McGonigle, and A. E. Russell (2004). “Lost at Sea: Where Is All the Plastic?” In: *Science* 304.5672. DOI: [10.1126/science.1094559](https://doi.org/10.1126/science.1094559).
- Ueda, K., Y. Kameda, E. Fujita, S. Rachi, Y. Iwasaki, R. Tai, and W. Naito (2025). “Concentrations and characteristics of microplastic particles collected by neuston net or pump system in the surface layer of Tokyo Bay”. In: *Region. Stud. Mar. Sci.* 84. DOI: [10.1016/j.rsma.2025.104108](https://doi.org/10.1016/j.rsma.2025.104108).
- Virkar, Y. and A. Clauset (2014). “POWER-LAW DISTRIBUTIONS IN BINNED EMPIRICAL DATA”. In: *The Annals of Applied Statistics* 8.1. DOI: [10.1214/13-aos710](https://doi.org/10.1214/13-aos710).
- Yang, S., X. Lu, and X Wang (2024). “A Perspective on the Controversy over Global Emission Fluxes of Microplastics from Ocean into the Atmosphere”. In: *Environ. Sci. Technol.* 58.28. DOI: [10.1126/science.adl2746](https://doi.org/10.1126/science.adl2746).
- Zhao, S., T. J. Mincer, L. Lebreton, and M. Egger (2023). “Pelagic microplastics in the North Pacific Subtropical Gyre: A prevalent anthropogenic component of the particulate organic carbon pool”. In: *PNAS Nexus* 2.3. DOI: [10.1093/pnasnexus/pgad070](https://doi.org/10.1093/pnasnexus/pgad070).
- Zhao, S., Z. Zhang, L. Chen, Q. Cui, Y. Cui, D. Song, and L. Fang (2022). “Review on migration, transformation and ecological impacts of microplastics in soil”. In: *Applied Soil Ecology* 176. DOI: [10.1016/j.apsoil.2022.104486](https://doi.org/10.1016/j.apsoil.2022.104486).
- Zhu, X., W. Huang, M. Fang, Z. Liao, Y. Wang, L. Xu, Q. Mu, C. Shi, C. Lu, H. Deng, R. Dahlgren, and X. Shang (2021). “Airborne Microplastic Concentrations in Five Megacities of Northern and Southeast China”. In: *Environ. Sci. Technol* 55.19. DOI: [10.1021/acs.est.1c03618](https://doi.org/10.1021/acs.est.1c03618).

Supplementary information

Three discussed cases of published observational studies (Tekman et al. (2020), Zhu et al. (2021) and Zhao, Mincer, et al. (2023)) were discussed in Results and Discussion section. The results presented in Figures 10 and 11 were obtained using the following bins and the corresponding linear regressions:

